## SECTION-I

**Single Correct Answer Type** **15 Q. [3 M (–1)]**

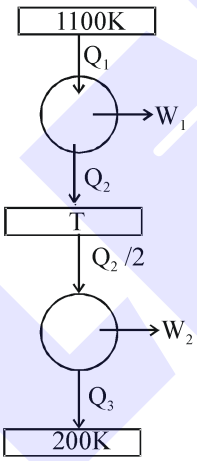
**1.** Diatomic gas is used in carnot heat engine. If efficiency of given carnot heat engine is 80%, then find the ratio of initial volume to final volume of gas during adiabatic expansion.

(A) $\left(\frac{1}{5}\right)^{3/2}$ (B) $\left(\frac{1}{3}\right)^{5/2}$ (C) $\left(\frac{1}{5}\right)^{5/2}$ (D) $\left(\frac{1}{5}\right)^{2/5}$

**2.** A carnot refrigerator operates between –3°C and 27°C. How much work must be done so as to extract 1 kJ of heat from the refrigerator?

(A) 100 J (B) $10^4$ J (C) $\frac{1000}{9}$ J (D) 900 J

**3.** Consider the given series combination of carnot cycles. If $W_1 = W_2$ then the value of T is (all temperatures are maintained at their respective values)




(A) 600 K (B) 800 K (C) 700 K (D) 900 K

**4.** In a refrigerator one removes heat from a lower temperature and deposits to the surroundings at a higher temperature. In this process, mechanical work has to be done, which is provided by an electric motor. If the motor is of 1kW power, and heat is transferred from –3°C to 27°C, find the heat taken out of the refrigerator per second assuming its efficiency is 50% of a perfect engine.

(A) 500 J (B) 5 kJ (C) 12 kJ (D) 19 kJ

**5.** A household refrigerator with a coefficient of performance 1.2 removes heat from the refrigerated space at the rate of 60kJ/min.What would be cost of running this fridge for one month (30 days) (assuming each day it is used for 4 hours and cost of one electrical unit is 6Rs.):-

(A) 180 Rs. (B) 300 Rs. (C) 480Rs. (D) 600 Rs.

**6.** On doubling the temperature of source, the efficiency of a heat engine becomes triple. The new efficiency is :

(A) 10% (B) 20% (C) 60% (D) 45%

**7.** A carnot refrigerator has a coefficient of performance as 5. If the same cycle is carried out in reverse, it will behave like a heat engine. What is the efficiency of this heat engine?

(A) 20% (B) 16.67% (C) 28.33% (D) 40%

**8.** Which of the following is a FALSE statement?

(A) The difference in entropy between two states of a system is independent of the path between the two states.

(B) Entropy is a quantitative measure of disorder.

(C) The total entropy change in one cycle of a Carnot engine is zero.

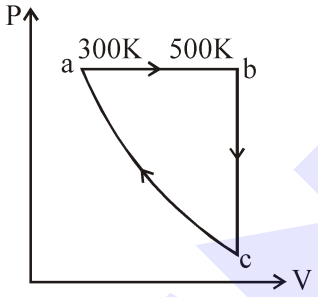
(D) The entropy of an isolated system can decrease.



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**9.** Two Carnot engines A and B are operated in succession. The first one, A receives heat from a source at $T_1 = 800$ K and rejects to sink at $T_2$K. The second engine B receives heat rejected by the first engine and rejects to another sink at $T_3 = 300$K If the work outputs of two engines are equal, then the value of $T_2$ is :-
(A) 100K (B) 300K (C) 550K (D) 700K

**10.** A heat engine performs the reversible cycle abca with 9.0 moles of an ideal gas. Path ca is an adiabatic process. The temperature at point a and b are 300 K and 500 K, respectively. The adiabatic constant of the gas is 1.60. In figure, the heat absorbed by the gas in path ab, in kJ, is $\left(R = \frac{25}{3}\right)$ :

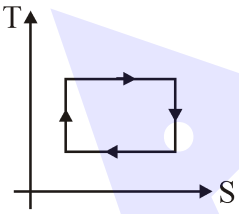


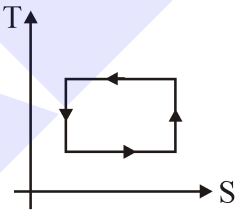

(A) Zero (B) + 25 (C) + 40 (D) – 25

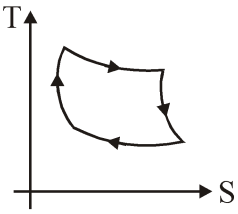
**11.** "Heat cannot be itself flow from a body at lower temperature to a body at higher temperature" is a statement or consequence of- **[AIEEE - 2003]**
(A) second law of thermodynamics (B) conservation of momentum
(C) conservation of mass (D) first law of thermodynamics

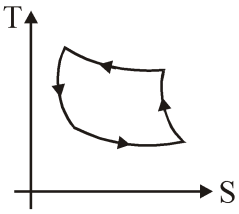
**12.** Which of the following process is reversible :
(A) A block moving on a rough surface comes to rest after same time.
(B) Two gases are mixed in a container
(C) A block kept on a smooth surface is connected to a fixed spring and is oscillating horizontally.
(D) The water is heated on a stove.

**13.** The carnot cycle for a heat engine on TS diagram is

(A) 

(B) 

(C) 

(D) 


**14.** An ideal gas is taken from temperature $T_i$ to $T_f$. $T_f > T_i$ by 3 different reversible processes.
(a) isobaric (b) isochoric (c) adiabatic
The change in entropy is
$\Delta S_b$, $\Delta S_u$ & $\Delta S_S$ respectively. Choose the correct relation between them.
(A) $\Delta S_b > \Delta S_u > \Delta S_S$ (B) $\Delta S_S > \Delta S_u > \Delta S_b$
(C) $\Delta S_b > \Delta S_S > \Delta S_u$ (D) $\Delta S_u > \Delta S_b > \Delta S_S$

**15.** According to the second law of thermodynamics:
(A) Heat energy cannot be completely converted to work
(B) Work cannot be completely converted to heat energy
(C) For all cyclic processes we have $dQ/T < 0$
(D) The reason all heat engine efficiencies are less than 100% is friction, which is unavoidable
(E) All of the above are true


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**Linked Comprehension Type** **(1 Para × 2Q.) [3 M (-1)]**
**(Single Correct Answer Type)**

**Paragraph for Question no. 16 & 17**

An air conditioner installed in a room is operated by carnot cycle in reverse. It acts like a carnot refrigerator. The room is closed from all sides. The walls lose heat to surrounding by conduction. The inside of room is at 27°C when outside is at 37°C. The AC consumes a power of 1000W in this case.

**16.** If the outside temperature increases to 47°C, what will be the power consumed if the inside temperature is still 27°C.

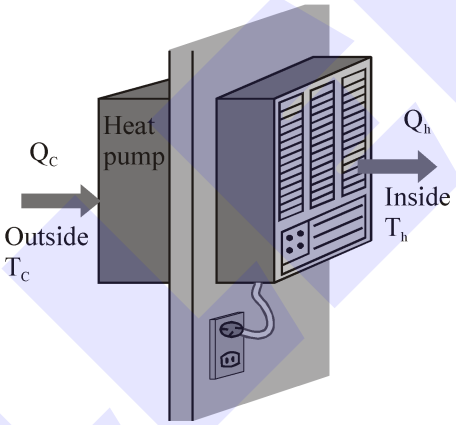
(A) 2000W (B) 3000W (C) 4000W (D) 8000W

**17.** If the outside temperature increases to 47°C, but the AC still operates at 1000W, what will be the inside temperature (approx.)

## SECTION-III

**Numerical Grid Type (Ranging from 0 to 9)** **1 Q. [4 M (0)]**

**18.** A heat pump used for heating shown in figure is essentially an air conditioner installed backward. It extracts energy from colder air outside and deposits it in a warmer room. Suppose the ratio of the actual energy entering the room to the work done by the device's motor is 20.0% of the theoretical maximum ratio. Determine the energy entering the room per joule of work done by the motor, given that the inside temperature is 27.0°C and the outside temperature is –3°C.



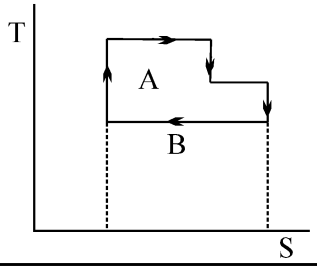

**Subjective Type** **12 Q. [4 M (0)]**

**19.** An ideal (Carnot) freezer in a kitchen has a constant temperature of 260 K, whereas the air in the kitchen has a constant temperature of 300 K. Suppose the insulation for the freezer is not perfect and conducts energy into the freezer at a rate of 0.150 W. Determine the average power required for the freezer's motor to maintain the constant temperature in the freezer.

**20.** A quantity of 0.20 mol of argon gas contained in a volume of 5.0 L mixes with 0.50 mol of neon gas contained in a volume of 5L, making a total volume of 10 L. Both gases are at the same temperature and can be regarded as ideal. Each volume is thermally isolated. What is the change in entropy ?

**21.** A reversible heat engine has an efficiency of 33.3%, removing heat from a hot reservoir and rejecting heat to a cold reservoir at 0°C. If the engine now operates in reverse, how long would it take of freeze 1.0 kg of water at 0°C, if it operates on a power of 186 W?

**22.** An engine is represented by a cyclic process shown in the T-S (temperature versus entropy)diagram. Here A represents the area of the region enclosed by the cycle and B of the area below the curve. Compute the efficiency cycle.

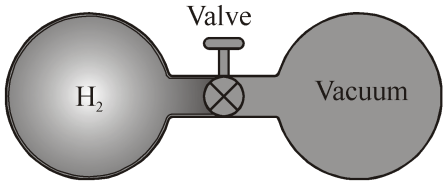




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**23.** Find the change in entropy during the transition of 8 grammes of oxygen from a volume of 10 litres at a temperature of 80°C to a volume of 40 litres at a temperature of 300°C.

**24.** A 1.00 mol sample of $H_2$ gas is contained in the left side of the container shown in figure, which has equal volumes left and right. The right side is evacuated. When the valve is opened, the gas streams into the right side. What is the final entropy change of the gas? Assume the container is so large that the hydrogen behaves as an ideal gas.




**25.** An ice tray contains 500 g of liquid water at 0°C. Calculate the change in entropy of the water as it freezes slowly and completely at 0°C.

**26.** A carnot engine working between 400 K and 800 K has a work output of 1200 J per cycle. What is the amount of heat energy supplied to the engine from source per cycle?

**27.** A carnot engine works as a refrigrator between 250 K and 300 K. If it receives 750 cal of heat from the reservoir at the lower temperature. Calculate the amount of heat rejected at the higher temperature.

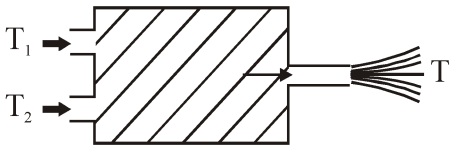
**28.** A refrigerator takes heat from water at 0°C and transfer it to room at 27°C. If 100 kg of water is converted in ice at 0°C then calculate the workdone. (Latent heat of ice is $3.4 \times 10^5$ J/kg)

**29.** An electric freezer is turned on inside a tent for a long time. It is 0 °C outside the tent, +1 °C inside the tent, and –13 °C inside the freezer. What would be the equilibrium temperature inside the tent if another identical freezer is turned on inside the tent? The outside temperature remains the same. The freezers are identical and follow the Carnot cycle.

**30.** A self-contained machine only inputs two equal steady streams of hot and cold water at temperatures $T_1$ and $T_2$. Its only output is a single high-speed jet of water. The heat capacity per unit mass of water, C, may be assumed to be independent of temperature. The machine is in a steady state and the kinetic energy in the incoming streams is negligible.

(a) What is the speed of the jet in terms of $T_1$, $T_2$ and T, where T is the temperature of water in the jet?

(b) What is the maximum possible speed of the jet?




**GR # Thermodynaimcs Second Law & Entropy** **ANSWER KEY**

## SECTION-I

**Single Correct Answer Type** **15 Q. [3 M (–1)]**

| | | | | |
|---|---|---|---|---|
| **1. Ans. (C)** | **2. Ans. (C)** | **3. Ans. (B)** | **4. Ans. (D)** | **5. Ans. (D)** |
| **6. Ans. (C)** | **7. Ans. (B)** | **8. Ans. (D)** | **9. Ans. (C)** | **10. Ans. (C)** |
| **11. Ans. (A)** | **12. Ans. (C)** | **13. Ans. (A)** | **14. Ans. (A)** | **15. Ans. (A)** |

**Linked Comprehension Type** **(1 Para × 2Q.) [3 M (-1)]**

**(Single Correct Answer Type)**

**16. Ans. (C)** **17. Ans. (D)**

## SECTION-III

**Numerical Grid Type (Ranging from 0 to 9)** **1 Q. [4 M (0)]**

**18. Ans. 2**

**Subjective Type** **12 Q. [4 M (0)]**

**19. Ans.** $\frac{3}{130}$W **20. Ans.** Approx. 4 J/K **21. Ans.** 15 min **22. Ans.** A / (A + B)

**23. Ans.** 5.4 J/k **24. Ans.** 5.76 J/k **25. Ans.** –610 J/k **26. Ans.** 2400 joule per cycle

**27. Ans.** 900 cal **28. Ans.** $3.36 \times 10^6$ J **29. Ans.** 1.88°C

**30. Ans.** (a) $v = \sqrt{c(T_1 + T_2 - 2T)}$ (b) $v_{max} = \sqrt{c}\left(\sqrt{T_1} - \sqrt{T_2}\right)$



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